ENVIRONMENTAL SCIENCE

An Environmental science course provides an interdisciplinary foundation in ecology, biology, chemistry, and policy, focusing on sustainability, biodiversity, pollution management, and natural resource conservation. Core topics include ecosystem structure, climate change, energy systems, and environmental regulations, designed to equip students with analytical skills to solve complex environmental challenges.

Typical Course Curriculum

- **Introduction to Environmental Science**
  - Fundamental principles
  - Environmental history
  - Earth systems (atmosphere, biosphere, hydrosphere, geosphere), and sustainability

- **Ecosystems and Biodiversity**
  - Structure and function
  - Energy flow
  - Food webs
  - Ecological succession
  - Conservation biology

- ## Natural Resource Management
  - Renewable and non-renewable resources
  - Forest management
  - Water resources
  - Mineral extraction, and
  - Agriculture

- ## Pollution and Environmental Health
  - Types of pollution (water, air and soil)
  - Waste management
  - Toxic contaminants, and
  - Environmental impact assessments.

- ## Climate change and Policy
    - Global Climate feedbacks
    - Renewable energy sources
    - Sustainable development, and
    - Environmental law

- ## Research Methods and Technology
    - Laboratory techniques
    - Data analysis
    - Geographic Information Systems (GIS), and
    - Remote sensing

Key Learning Outcomes

- Analyze ecological processes and human impacts on the environment
- Evaluate sustainability strategies at local, national, and global levels
- Apply scientific methods and laboratory techniques to environmental research
    - Develop skills in environmental policy analysis and management

Typical Course curriculum

- **Duration:** At one week, two weeks, one month, three months, nine months, one year modular courses
- **Assessment:** Mix of lectures, laboratory experiments, field studies, research projects, and examinations
- **Requirements:** Background in biology, chemistry, mathematics, or geography.